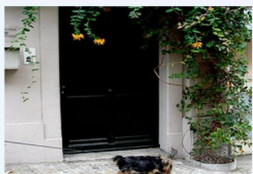


E8: Gemma4-Guided Multi-Dimensional Query Rewriting + MagicLens Hybrid Retrieval

query image + question -> 5 retrieval dimensions -> 5x5 MagicLens evidence pool -> RRF Top-5 -> fixed VLM answer

(a) Query + planning

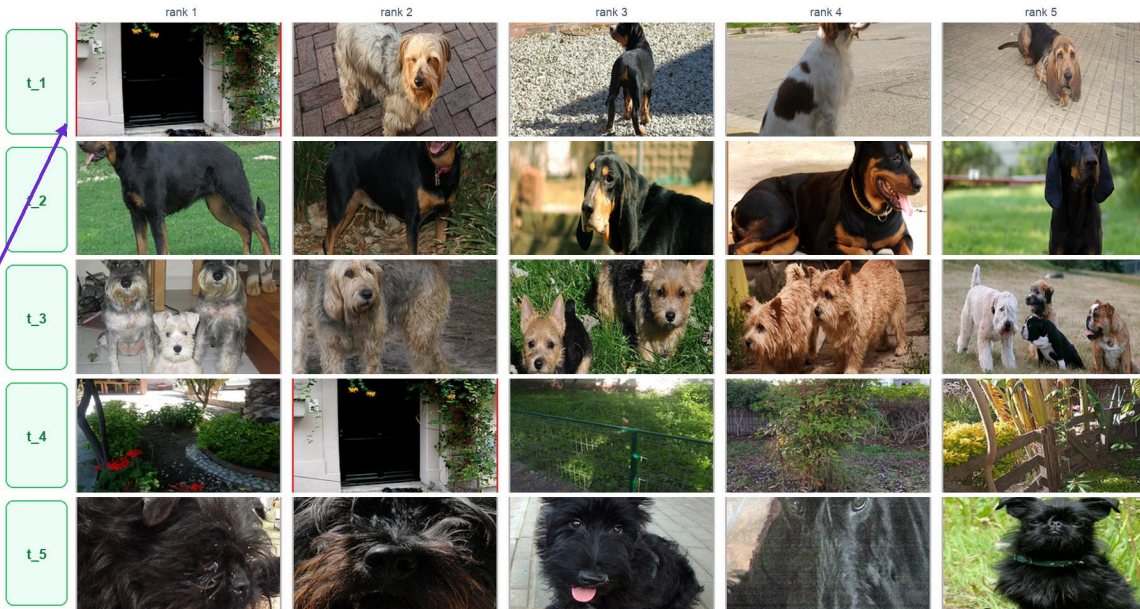


Question: Can you identify this animal?
Choices: A/B/C/D

Gemma4 Planner
image + question + choices
-> 5 retrieval dimensions

- 1 dog sitting on cobblestone path in front of a building
- 2 black and tan dog breed identification
- 3 terrier dog breeds comparison
- 4 outdoor scene with ivy and stone pathway
- 5 close-up of a small black dog

(b) Per-dimension MagicLens retrieval: 5 dimensions x Top-5



(c) RRF evidence fusion and fixed VLM answering

RRF Fusion
sum $1/(60 \cdot \text{rank})$
select Top-5

Gemma4
Evidence
Descriptions

Final RRF Evidence Set (Top-5 candidates)



LLaVA-OneVision
fixed VLM
A/B/C/D

E8 full result
1353 samples
56.32% Acc
+8.35 vs E7